LOW ACRYLONITRILE NBR PATENT RESEARCH REPORT

1. ABSTRACT

This report summarizes the findings from the analysis of patent US9926439 related to Low Acrylonitrile NBR. The patent describes a nitrile rubber article having natural rubber characteristics. The analysis focused on the general parameters and examples provided in the patent, including elongation, DPG, ZMBT, and ZDEC. A total of 1 primary patent was analyzed, with no competitor patents or website sources found. The report highlights the technical approaches and trends observed in the patent evidence.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: US9926439 - Patent Title: Nitrile rubber article having natural rubber characteristics - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2018-03-27 - Relevance to target: The patent is relevant to Low Acrylonitrile NBR as it describes a nitrile rubber article with improved properties.

Technical Method - Elongation: 400 % (The amount of force at 400% elongation required to achieve the same level of stretching in the eight comparative examples) - Dpg: 0.5 phr (The compounds are diphenyl guanidine (DPG), zinc mercaptobenzothizole (ZMBT), and zincdiethyldithiocarbamate (ZDEC)) - Zmbt: 0.25 phr (The compounds are diphenyl guanidine (DPG), zinc mercaptobenzothizole (ZMBT), and zincdiethyldithiocarbamate (ZDEC)) - Zdec: 0.25 phr (The compounds are diphenyl guanidine (DPG), zinc mercaptobenzothizole (ZMBT), and zincdiethyldithiocarbamate (ZDEC))

Relevant Experimental Evidence - The patent provides examples of nitrile rubber articles with improved properties, including Example 12. - The examples demonstrate the use of DPG, ZMBT, and ZDEC in the production of nitrile rubber articles. - The patent highlights the importance of controlling the amount of force at 400% elongation to achieve the desired level of stretching.

Technical Relevance This patent is relevant to Low Acrylonitrile NBR as it describes a nitrile rubber article with improved properties, which could be useful for applications requiring low acrylonitrile content.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Trend Analysis: - Insufficient comparable evidence was available to identify a reliable cross-patent trend.

4. CONCLUSION

The analysis of patent US9926439 highlights the importance of controlling the amount of force at 400% elongation and the use of DPG, ZMBT, and ZDEC in the production of nitrile rubber articles with improved properties. The patent is relevant to Low Acrylonitrile NBR and provides useful information for the development of nitrile rubber articles with low acrylonitrile content.

5. REFERENCES

- US9926439